Policy Claim Classification via LLM Fine-Tuning

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Problem & Motivation

Why classify policy claims automatically?

70+

countries have published AI strategies as of 2024

1000s

of policy claims scattered across lengthy documents

5

distinct policy domains require systematic tracking

Manual classification is slow, inconsistent, and does not scale.
Can a fine-tuned LLM do it reliably?

Task Definition

Single-label text classification (5 classes) via generative classifier

Input *"5G network coverage reached 57% of the global population in 2024 but significant gaps remain in developing countries."*

Output **Infrastructure**

Model generates the label as text — not a classification head

Strategy	Regulation	Infrastructure	Capacity	Inclusion
Direction, coordination	Laws, rules, enforcement	Networks, hardware	Education, skills	Equity for disadvantaged

Literature Review

1 LoRA Land: 310 Fine-tuned LLMs that Rival GPT-4

Zhao et al. (2024) • arXiv:2405.00732

Key finding: QLoRA fine-tuning of 8B models outperformed GPT-4 on 85% of tasks, with 25–50% average improvement. Same setup as ours: 4-bit quantization, LoRA rank 8, cosine scheduler.

2 Leveraging LLMs for Topic Classification in Public Affairs

Peña et al. (2023) • arXiv:2306.02864

Key finding: LLMs effectively classify 33K+ public affairs documents. Validates that automated policy text classification is a meaningful and active research area.

Dataset

7 Source Documents

- OECD (2021) — National AI Strategies
- World Bank (2025) — Digital Progress Report
- World Bank (2025) — Digital Public Infrastructure
- UNDP (2024) — Digital Inclusion Playbook
- UNESCO (2021) — Ethics of AI
- World Bank (2024) — AI Governance Trends
- EY (2024) — EU AI Act Summary

193

total claims

Train: 143 | Test: 50 (stratified 80/20)

Labeling Decision Tree

Step 1	Step 2	Step 3	Step 4	Step 5
Disadvantaged group named?	Law/regulation is core action?	Build/deploy system?	Education/ training?	None of above
→ **Inclusion**	→ **Regulation**	→ **Infrastructure**	→ **Capacity**	→ **Strategy**

Model & Fine-Tuning Setup

Parameter	Value	Why
Base model	Llama 3.1 8B Instruct	Best ecosystem & community support for beginners
Quantization	4-bit NF4 (bitsandbytes)	16 GB → 5.4 GB — fits T4 16 GB VRAM
Adaptation	QLoRA (rank 16, alpha 32)	Only ~0.5% of params trained
Training	3 epochs, batch 8, LR 2e-4	Prevents overfitting on 143 samples
Max seq length	256 tokens	Policy claims avg ~80 tokens; saves 60% VRAM
Library	Unsloth	2× faster training, 60% less memory

~5 min

Training time
on T4 GPU

~30 MB

Adapter size
(vs 16 GB full model)

0.5%

Trainable params
(LoRA efficiency)

Evaluation Design

Metrics

Tier	Metric	What It Measures
Primary	Accuracy	Overall correct predictions out of 50
Primary	Macro F1	Average per-class F1 (equal weight)
Secondary	Per-class F1	Pinpoints strong/weak categories
Secondary	Confusion Matrix	Which pairs get confused
Secondary	Invalid Rate	% of non-label outputs

Data Leakage Prevention

- ✓ Stratified 80/20 split (min 10 per class in test)
- ✓ Test set NEVER used during training
- ✓ Same test set for both baseline and fine-tuned
- ✓ Identical prompt template for fair comparison

Results: Baseline vs Fine-Tuned

Metric	Baseline (zero-shot)	Fine-Tuned (QLoRA)	Δ Change
Accuracy	83.7%	**94.0%**	**+10.3%p**
Macro F1	0.838	**0.941**	**+0.103**
Invalid Rate	14.0%	**0.0%**	**−14.0%p**

Per-Class F1

Category	Baseline	Fine-Tuned	Δ
Strategy	0.67	**0.95**	+0.28
Regulation	0.95	**0.95**	0.00
Infrastructure	0.75	**0.91**	+0.16
Capacity	0.82	**0.95**	+0.13
Inclusion	1.00	**0.95**	−0.05

Key Insight

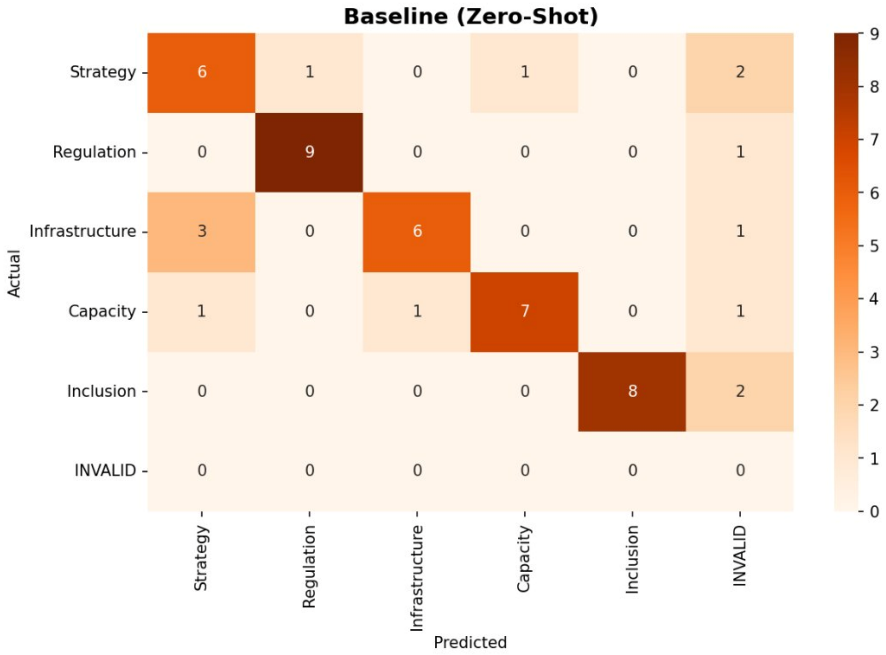
Strategy improved most (+0.28)

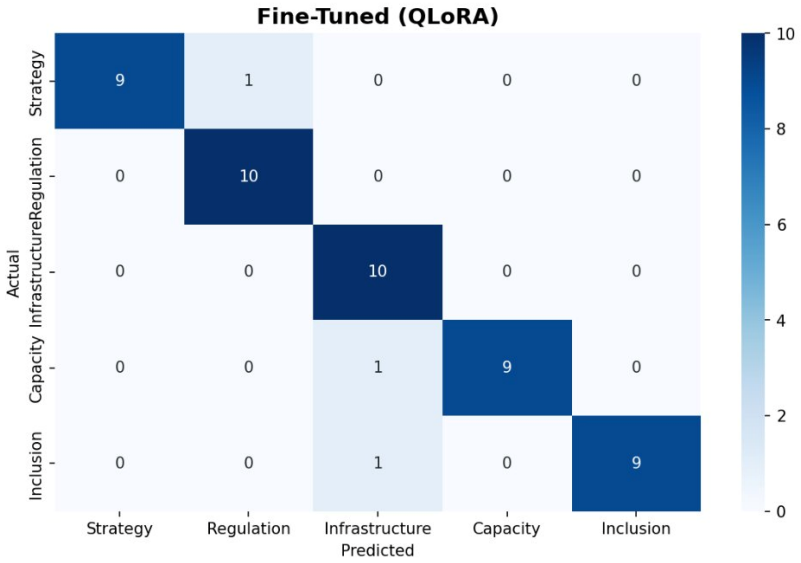
Baseline confused Strategy with other categories and produced 14% invalid outputs.

Fine-tuning eliminated all invalid predictions.

Results: Visual Comparison

Confusion Matrix: Baseline vs Fine-Tuned

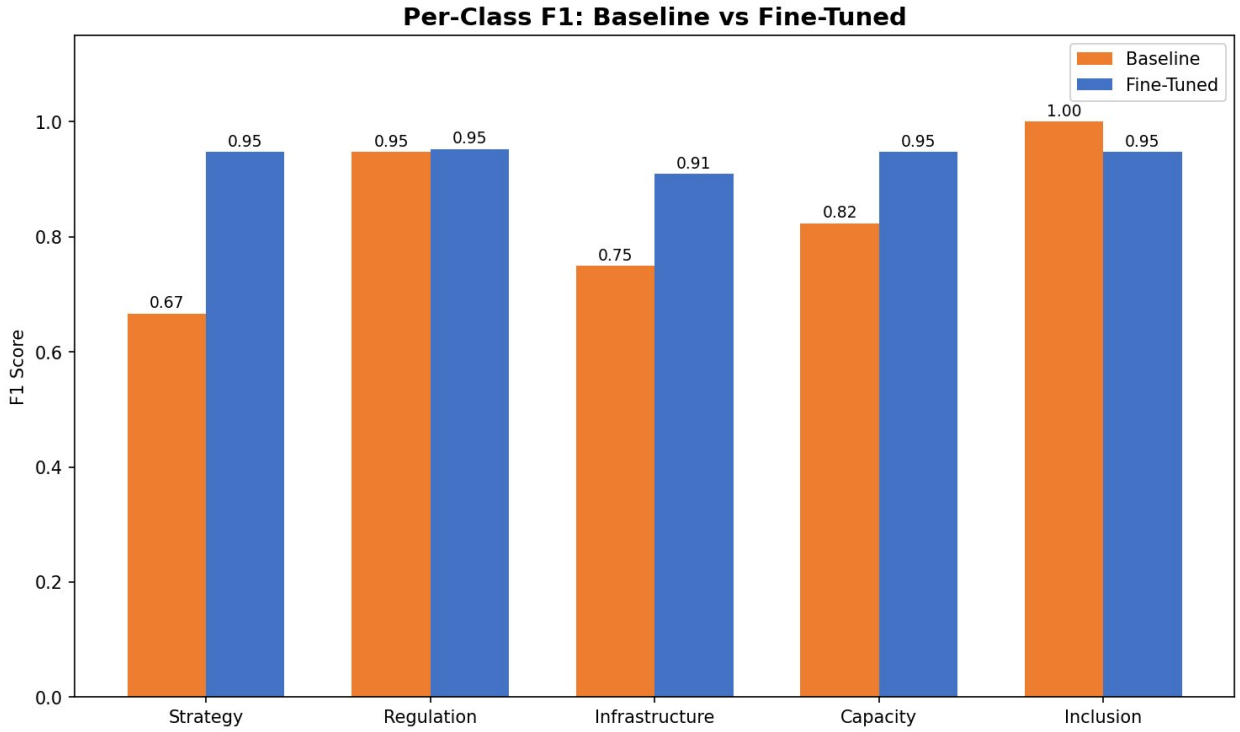




Left: Baseline errors spread across categories + 7 INVALID

Right: Fine-tuned errors concentrated, 0 INVALID

Results: Per-Class F1 Comparison



Takeaways

Biggest gain:
Strategy +0.28

Most stable:
Regulation 0.95 → 0.95

Slight regression:
Inclusion 1.00 → 0.95
(1 boundary error)

All 5 categories now above 0.90 F1 — baseline had 3 below 0.85

Error Analysis

What Improved

Strategy F1: 0.67 → 0.95
Baseline confused Strategy with Capacity and produced INVALID outputs

Invalid Rate: 14% → 0%
Fine-tuning taught the model to always output a valid label

What's Still Hard

Inclusion F1: 1.00 → 0.95
1 Inclusion claim misclassified as Infrastructure — boundary overlap

Infrastructure remains lowest (0.91)
Overlaps with Inclusion when infrastructure serves disadvantaged groups

Confusion Matrix Insight

Baseline: Errors spread across many category pairs (Strategy→Capacity, Infrastructure→Strategy, Inclusion→INVALID). 7 INVALID outputs.

Fine-Tuned: Only 3 errors remain: Strategy→Regulation (1), Capacity→Infrastructure (1), Inclusion→Infrastructure (1). Zero INVALID outputs.

Limitations & Next Steps

Limitations

Small dataset
193 claims (143 train) — results may be split-sensitive

Single annotator
No inter-annotator agreement measured

English only
All 7 sources are from international orgs in English

Source homogeneity
May not generalize to national policy documents

Next Steps

Scale to 500+ claims
Include national strategies and think-tank reports

Multi-annotator validation
Measure Cohen's Kappa on subset

Compare model architectures
Qwen 2.5 7B, Gemma 2 9B

Hyperparameter search
LoRA rank (8/16/32), LR, epochs

End of Document